

Repository Quality Scoring Guide (Current)

This guide summarizes how repository code-quality scores are computed in `backend/services/code_quality_analyzer.py` and surfaced in the report via `backend/services/report_generator.py`.

For a full, implementation-level explanation (including worked examples, optional dependency behavior, and dynamic calibration), see `docs/scoring-deep-dive.md`.

What the Report/Frontend Shows

The dashboard reads `report.codeQualityReport.categoryScores` (portfolio-level averages across analyzed repositories):

- `documentation`
- `codeOrganization`
- `commitQuality`
- `codeHealth` (only when Code Health is enabled and available)

Notes:

- LICENSE/tests/CI-CD signals are intentionally excluded from scoring in the current analyzer.
- Code Health is optional and may be unavailable depending on repo contents and installed libraries.

Score bounds

All component scores are clamped to `0..100`.

Category calculations (per repository)

1) Documentation (0–100)

Signals:

- README quality score $\times 0.65$
- `+18` if a docs folder exists at repo root (`docs/`, `doc/`, ...)
- `+10` if an architecture/design doc exists at root (e.g., `architecture.md`)
- `+10` if an examples/demo folder exists at root
- `+7` if README includes a usage signal
- `+10` if `CONTRIBUTING.md` exists at root

2) Code Organization (0–100)

Formula:

- Base by structure label: `excellent=100`, `good=75`, `basic=50`, `minimal=25`
- Richness bonus: `min(15, directory_count * 1.5)`
- Final: `base * 0.85 + richness_bonus`

3) Commit Quality (0–100)

Formula:

- $0.45 * \text{commit_frequency_score} + 0.55 * \text{commit_message_score}$

Commit message quality includes an anti-repetition penalty when RapidFuzz is available.

4) Code Health (0–100, optional)

If enabled, this analyzes real source files (.py/.js/.ts/.java) using Radon/Esprima/Javalang when available.

Per-repo totals and portfolio totals

- Per-repo total: weighted average of the categories present (weights are re-normalized if optional categories are missing).
- Portfolio overall score: weighted average of repo scores, weighted by stars and recency.